

EXPERIMENT 22: PERFORM SHARPENING OF IMAGE USING LAPLACIAN MASK WITH POSITIVE CENTER COEFFICIENT

Aim

To sharpen an image using a Laplacian mask with a positive center coefficient in Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image to grayscale.
4. Define the positive-center Laplacian kernel.
5. Apply the kernel using filter2D().
6. Display and save the sharpened image.

Laplacian Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Python Code

```
import cv2
import numpy as np

img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.array([[0, -1, 0],
                   [-1, 5, -1],
```

```
[0, -1, 0]], dtype=np.float32)

sharpened = cv2.filter2D(gray, -1, kernel)

cv2.imwrite("laplacian_positive_output.jpg", sharpened)

cv2.imshow("Input Image", gray)
cv2.imshow("Sharpened Image", sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the image was successfully sharpened using the Laplacian mask with a positive center coefficient.